

```

//gp generator
module gp (i_a, i_b, o_g, o_p);
    input i_a, i_b;
    output o_g, o_p;

    assign o_g = i_a & i_b;
    assign o_p = i_a ^ i_b;
endmodule

// first cl unit
module cl1 (i_g, i_p, i_cin, o_s, o_gmerge, o_pmerge);
    input [2:0] i_g, i_p;
    input i_cin;
    output [2:0] o_s;
    output o_gmerge, o_pmerge;

    assign o_s[0] = i_p[0] ^ i_cin;
    assign o_s[1] = i_p[1] ^ (i_g[0] | i_p[0] & i_cin);
    assign o_s[2] = i_p[2] ^ (i_g[1] | i_p[1] & i_g[0] | i_p[1] & i_p[0] & i_cin);
    assign o_gmerge = i_g[2] | i_p[2] & i_g[1] | i_p[2] & i_p[1] & i_g[0];
    assign o_pmerge = i_p[2] & i_p[1] & i_p[0];
endmodule

// multi level cl unit
module clMult (i_g, i_p, i_cin, o_cout, o_gmerge, o_pmerge);
    input [2:0] i_g, i_p;
    input i_cin;
    output [1:0] o_cout;
    output o_gmerge, o_pmerge;

    assign o_cout[0] = i_g[0] | i_p[0] & i_cin;
    assign o_cout[1] = i_g[1] | i_p[1] & i_g[0] | i_p[1] & i_p[0] & i_cin;
    assign o_gmerge = i_g[2] | i_p[2] & i_g[1] | i_p[2] & i_p[1] & i_g[0];
    assign o_pmerge = i_p[2] & i_p[1] & i_p[0];
endmodule

// CLA module
module CLA27 (i_a, i_b, i_cin, o_s, o_cout);
    input [26:0] i_a, i_b;
    input i_cin;
    output [26:0] o_s;
    output o_cout;

    // generate g & p
    wire [26:0] w_g;
    wire [26:0] w_p;

```

```

gp A0 (.i_a(i_a[0]), .i_b(i_b[0]), .o_g(w_g[0]), .o_p(w_p[0]));
gp A1 (.i_a(i_a[1]), .i_b(i_b[1]), .o_g(w_g[1]), .o_p(w_p[1]));
gp A2 (.i_a(i_a[2]), .i_b(i_b[2]), .o_g(w_g[2]), .o_p(w_p[2]));
gp A3 (.i_a(i_a[3]), .i_b(i_b[3]), .o_g(w_g[3]), .o_p(w_p[3]));
gp A4 (.i_a(i_a[4]), .i_b(i_b[4]), .o_g(w_g[4]), .o_p(w_p[4]));
gp A5 (.i_a(i_a[5]), .i_b(i_b[5]), .o_g(w_g[5]), .o_p(w_p[5]));
gp A6 (.i_a(i_a[6]), .i_b(i_b[6]), .o_g(w_g[6]), .o_p(w_p[6]));
gp A7 (.i_a(i_a[7]), .i_b(i_b[7]), .o_g(w_g[7]), .o_p(w_p[7]));
gp A8 (.i_a(i_a[8]), .i_b(i_b[8]), .o_g(w_g[8]), .o_p(w_p[8]));
gp A9 (.i_a(i_a[9]), .i_b(i_b[9]), .o_g(w_g[9]), .o_p(w_p[9]));
gp A10 (.i_a(i_a[10]), .i_b(i_b[10]), .o_g(w_g[10]), .o_p(w_p[10]));
gp A11 (.i_a(i_a[11]), .i_b(i_b[11]), .o_g(w_g[11]), .o_p(w_p[11]));
gp A12 (.i_a(i_a[12]), .i_b(i_b[12]), .o_g(w_g[12]), .o_p(w_p[12]));
gp A13 (.i_a(i_a[13]), .i_b(i_b[13]), .o_g(w_g[13]), .o_p(w_p[13]));
gp A14 (.i_a(i_a[14]), .i_b(i_b[14]), .o_g(w_g[14]), .o_p(w_p[14]));
gp A15 (.i_a(i_a[15]), .i_b(i_b[15]), .o_g(w_g[15]), .o_p(w_p[15]));
gp A16 (.i_a(i_a[16]), .i_b(i_b[16]), .o_g(w_g[16]), .o_p(w_p[16]));
gp A17 (.i_a(i_a[17]), .i_b(i_b[17]), .o_g(w_g[17]), .o_p(w_p[17]));
gp A18 (.i_a(i_a[18]), .i_b(i_b[18]), .o_g(w_g[18]), .o_p(w_p[18]));
gp A19 (.i_a(i_a[19]), .i_b(i_b[19]), .o_g(w_g[19]), .o_p(w_p[19]));
gp A20 (.i_a(i_a[20]), .i_b(i_b[20]), .o_g(w_g[20]), .o_p(w_p[20]));
gp A21 (.i_a(i_a[21]), .i_b(i_b[21]), .o_g(w_g[21]), .o_p(w_p[21]));
gp A22 (.i_a(i_a[22]), .i_b(i_b[22]), .o_g(w_g[22]), .o_p(w_p[22]));
gp A23 (.i_a(i_a[23]), .i_b(i_b[23]), .o_g(w_g[23]), .o_p(w_p[23]));
gp A24 (.i_a(i_a[24]), .i_b(i_b[24]), .o_g(w_g[24]), .o_p(w_p[24]));
gp A25 (.i_a(i_a[25]), .i_b(i_b[25]), .o_g(w_g[25]), .o_p(w_p[25]));
gp A26 (.i_a(i_a[26]), .i_b(i_b[26]), .o_g(w_g[26]), .o_p(w_p[26]));

```

//first layer of carry lookahead unit

```

wire [7:0]w_c;
wire [12:0]w_gmerged;
wire [12:0]w_pmerged;

```

```

cl1 U0 (.i_g(w_g[2:0]), .i_p(w_p[2:0]), .i_cin(i_cin), .o_s(o_s[2:0]),
.o_gmerge(w_gmerged[0]), .o_pmerge(w_pmerged[0]));
cl1 U1 (.i_g(w_g[5:3]), .i_p(w_p[5:3]), .i_cin(w_c[0]), .o_s(o_s[5:3]),
.o_gmerge(w_gmerged[1]), .o_pmerge(w_pmerged[1]));
cl1 U2 (.i_g(w_g[8:6]), .i_p(w_p[8:6]), .i_cin(w_c[1]), .o_s(o_s[8:6]),
.o_gmerge(w_gmerged[2]), .o_pmerge(w_pmerged[2]));
cl1 U3 (.i_g(w_g[11:9]), .i_p(w_p[11:9]), .i_cin(w_c[2]), .o_s(o_s[11:9]),
.o_gmerge(w_gmerged[3]), .o_pmerge(w_pmerged[3]));
cl1 U4 (.i_g(w_g[14:12]), .i_p(w_p[14:12]), .i_cin(w_c[3]), .o_s(o_s[14:12]),
.o_gmerge(w_gmerged[4]), .o_pmerge(w_pmerged[4]));
cl1 U5 (.i_g(w_g[17:15]), .i_p(w_p[17:15]), .i_cin(w_c[4]), .o_s(o_s[17:15]),
.o_gmerge(w_gmerged[5]), .o_pmerge(w_pmerged[5]));
cl1 U6 (.i_g(w_g[20:18]), .i_p(w_p[20:18]), .i_cin(w_c[5]), .o_s(o_s[20:18]),
.o_gmerge(w_gmerged[6]), .o_pmerge(w_pmerged[6]));

```

```

        cl1 U7 (.i_g(w_g[23:21]), .i_p(w_p[23:21]), .i_cin(w_c[6]), .o_s(o_s[23:21]),
.o_gmerge(w_gmerged[7]), .o_pmerge(w_pmerged[7]));
        cl1 U8 (.i_g(w_g[26:24]), .i_p(w_p[26:24]), .i_cin(w_c[7]), .o_s(o_s[26:24]),
.o_gmerge(w_gmerged[8]), .o_pmerge(w_pmerged[8]));

        // second layer
        clMult U9 (.i_g(w_gmerged[2:0]), .i_p(w_pmerged[2:0]), .i_cin(i_cin),
.o_cout(w_c[1:0]), .o_gmerge(w_gmerged[9]), .o_pmerge(w_pmerged[9]));
        clMult U10 (.i_g(w_gmerged[5:3]), .i_p(w_pmerged[5:3]), .i_cin(w_c[2]),
.o_cout(w_c[4:3]), .o_gmerge(w_gmerged[10]), .o_pmerge(w_pmerged[10]));
        clMult U11 (.i_g(w_gmerged[8:6]), .i_p(w_pmerged[8:6]), .i_cin(w_c[5]),
.o_cout(w_c[7:6]), .o_gmerge(w_gmerged[11]), .o_pmerge(w_pmerged[11]));

        // thrid layer
        wire [1:0] C_L3_OUT_BITS;
        clMult U12 (.i_g(w_gmerged[11:9]), .i_p(w_pmerged[11:9]), .i_cin(i_cin),
.o_cout(C_L3_OUT_BITS[1:0]), .o_gmerge(w_gmerged[12]), .o_pmerge(w_pmerged[12]));

        assign w_c[2] = C_L3_OUT_BITS[0];
        assign w_c[5] = C_L3_OUT_BITS[1];

        //calculating o_cout
        assign o_cout = w_gmerged[12] | w_pmerged[12] & i_cin;

endmodule

// 81 bit CLA
module CLA81 (i_a,i_b,i_cin, o_s, o_cout);
    input [80:0] i_a, i_b;
    input i_cin;
    output [80:0] o_s;
    output o_cout;

    wire [2:0]w_c;

    CLA27 U0 (.i_a(i_a[26:0]), .i_b(i_b[26:0]), .i_cin(i_cin), .o_s(o_s[26:0]),
.o_cout(w_c[0]));
    CLA27 U1 (.i_a(i_a[53:27]), .i_b(i_b[53:27]), .i_cin(w_c[0]), .o_s(o_s[53:27]),
.o_cout(w_c[1]));
    CLA27 U2 (.i_a(i_a[80:54]), .i_b(i_b[80:54]), .i_cin(w_c[1]), .o_s(o_s[80:54]),
.o_cout(o_cout));
endmodule

`timescale 1ns/1ns
module test_top();

```

```
reg [80:0] i_a, i_b;  
reg i_cin;
```

```
wire[80:0] o_s;  
wire o_cout;
```

```
// netlist  
CLA81 U0 (.i_a(i_a), .i_b(i_b), .i_cin(i_cin), .o_s(o_s), .o_cout(o_cout));
```

```
// simulation for the CLA 27
```

```
initial begin
```

```
    i_a <= 0;  
    i_b <= 0;  
    i_cin <= 1'b0;  
    //o_c:0  
    //o_s:0(decimail) = 0(bin) = 0(hex)
```

```
    #10  
    i_a <= 134217727;  
    i_b <= 1;  
    i_cin <= 1'b1;  
    // 1  
    // 1 = 1 = 1
```

```
    #10  
    i_a <= 134217727;  
    i_b <= 1;  
    i_cin <= 1'b0;  
    //1  
    // 0 = 0 = 0
```

```
    #10  
    i_a <= 123456789;  
    i_b <= 23456671;  
    i_cin <= 1'b1;  
    //1  
    //146913461 = 000110000011011100010110101 = C1B8B5
```

```
    #10  
    i_a <= 106;  
    i_b <= 79;  
    i_cin <= 1'b0;  
    //0  
    //185 = 10111001 = B9
```

```
    #10  
    i_a <= 123123123;  
    i_b <= 32132112;
```

```
i_cin <= 1'b1;  
//1  
//155255236 = 001010000010000000111000100 = 14101C4
```

```
#10  
i_a <= 11;  
i_b <= 13;  
i_cin <= 1'b0;  
//0  
//24 = 11000 = 18
```

```
#10  
i_a <= 191;  
i_b <= 33232231;  
i_cin <= 1'b1;  
//0  
//33232423 = 1111110110001011000100111 = 1FB1627
```

```
#10  
i_a <= 134217727;  
i_b <= 134217727;  
i_cin <= 0;  
//1  
//268435454 = 111111111111111111111110 = 7FFFFFFE
```

```
#10  
i_a <= 134217727; //7FFFFFFF  
i_b <= 134217727;  
i_cin <= 1;  
//1  
//268435455 = 111111111111111111111111 = 7FFFFFFF
```

```
end
```

```
endmodule
```

Test result

[illegible]